WEEK 6 · DAY 2 · INSTRUCTOR NOTES

When Will It Break? Remaining Useful Life + Quiz 4

For the trainer only.

The one idea for today: *remaining useful life = fit the trend on the past, extend it until it crosses a limit that is known up front, then say honestly how sure you are.* Honestly means three things: never let "no crossing" turn into a fake date, define "crossed" before you score a call, and report a window that says how far past the data you reached.

At a glance

Length	~2 hours, including Quiz 4 (20 min + 10 min marking)
Participants leave able to	fit a trend with `np.polyfit`, extend it to a threshold and turn the day number into a date; spot and fix the `np.argmax` "day 0" bug; check a model with a fuel-gauge sum; define a crossing three ways; backtest a remaining-useful-life method from several starting points; explain why a straight line calls a slowing fade early; write a planning figure with a range
Covers (original)	Week 6 Code Book: *Topic 6.4 Trend-to-threshold: remaining useful life* and the battery half of *Topic 6.5* (the 3-year call checked, and repeated from several origins); Week 6 quiz topic *trend-to-threshold and remaining useful life* (6 questions) plus Q29–30 from *backtesting*; Quiz 4 = the whole Week 6 MCQ file
Datasets	`week-06/data/degradation_assets_daily.csv` (1,825 days, 1 Jan 2021 – 30 Dec 2025, no gaps), `degradation_thresholds.csv`
Files you need open	animated slides, your **solution** notebook, `W6D2-quiz-4-answer-key.pdf`

The day before: checklist

☐ Run the **solution** notebook top to bottom: 13 of 13 ✅ (it takes a few seconds).

☐ Print `2-exercise/quiz/W6D2-quiz-4.pdf` (one per person), **or** import `W6D2-quiz-4-google-forms.csv` into Google Forms / your LMS.

☐ Print or open `4-instructor-only/W6D2-quiz-4-answer-key.pdf`. Don't share it until marking.

☐ Re-read slides 11–14 until you can tell the propellant story in one breath: *"the original printed the first day of the data; the line actually crosses after the search stopped; the fuel-gauge sum agrees."*

☐ Remind everyone that next session is **demo day**: five minutes each, with a working tool, a report and an honest test table.

Run of show (2 hours)

Time	Block	What happens	Slides
0:00	**Hook**	The danfo driver's fuel sum; NigComSat-1R's design life and the 2A/2B targets	1–4
0:10	**Three shapes**	Battery, amplifier, propellant and their limits; the propellant staircase	5–7
0:15	**Draw the line**	Trend-to-threshold; the two functions; the three calls; the argmax bug; solving the line; the fuel-gauge sum	8–14
0:27	**Check the call**	When did it cross (three answers); 13 vs 50 days early; backtest from 4 origins; why early; eclipse ageing; the long reach; golden rules	15–22
0:40	**Hands-on**	Parts A–C (45 min). Walk the room.	24
1:25	**Quiz 4**	20 minutes, phones down, no AI	23
1:45	**Mark + wrap**	Swap and mark, recap, demo-day reminder	25

If time is tight, skip slides 20 and 21 (eclipse ageing, long reach): Part C of the notebook covers both.

Explain it like this (in plain English)

- **Remaining useful life (RUL):** how long until a reading crosses its limit. The danfo driver does it every day: litres left ÷ litres per trip. A satellite engineer does the same with capacity, gain or fuel.
- **The limit is known up front.** It comes from engineering (`degradation_thresholds.csv`). What we forecast is *when* it's reached.
- **Trend-to-threshold:** `np.polyfit(x, y, 1)` gives a slope and an intercept. Extend that line into the future; the first day it's below the limit is the call.
- **Honest data rule:** the code book makes its calls with only the first three years (as if it were the morning of 1 Jan 2024), so the battery call can be checked against the two years that follow. Same idea as the Week 5 time split.
- **The bug:** `np.argmax` finds the first True. If there is no True at all it returns 0, so "never" becomes "day 0" = 1 Jan 2021. No error, no warning. Our `day_of_threshold` returns `None` instead.
- **Solving a straight line:** `value = slope × day + intercept` , so the day the value equals the limit is `(limit − intercept) / slope` . No search window needed.
- **The fuel-gauge sum** is the best bug detector there is: what's left above the reserve ÷ what's used a year. If it disagrees with the model, one of them is wrong.
- **"Crossed" has three honest meanings** for a noisy reading: the first day below, the first day the 30-day mean is below, the last day still at or above. Pick one *before* you score, and say which in the report.
- **Backtest the method, not just one call:** stand at the end of year 1, 2, 3 and 4 and make the call with what you had then. You learn how far off the method is, which way it errs (early or late), and how much warning it gives.
- **Why the straight line calls early here:** the battery's fade slows a little every year. A line from the faster early years overstates the later fade. For a battery, early is the safe side.

- **Long reach:** the amplifier call reaches about 9 years past its data. Report a planning figure (a range, how far it reaches, when you'll re-check), never a precise date.

The numbers (computed from the data)

The three parts (last reading on 30 Dec 2025, limit, 3-year line):

Part	Last reading	Limit	Slope of the 3-year line	3-year call
🔋 Battery capacity	77.84 %	80 %	−4.59 % a year	**1 May 2025** (486 days after the call on 1 Jan 2024)
📡 Amplifier gain	−0.783 dB	−2 dB	−0.162 dB a year	**5 May 2033** (9.3 years past the data it was fitted on)
⛽ Propellant	101.93 kg	5 kg	−9.72 kg a year	original prints **1 Jan 2021** (bug); fixed: no crossing before **13 Dec 2034** (end of the 4,000-day search); the line reaches 5 kg on **6 Dec 2035**

Propellant burns: 128 station-keeping burns in five years, one about every **14 days**, about **0.39 kg** each, about **26** a year. Five-year average use: **9.62 kg a year**.

Fuel-gauge sum: (101.93 − 5) = 96.9 kg ÷ 9.62 kg a year = **10.1 years** after 30 Dec 2025: about **January 2036**. The line says 6 Dec 2035. Two methods, one story.

When did the battery cross 80%?

Definition	Date	The 3-year call (1 May 2025) is…
① first day below 80%	**14 May 2025**	**13 days early** (the code book's answer, Quiz 4 Q29)
② first day the 30-day mean is below 80%	**20 Jun 2025**	**50 days early**
③ last day at or above 80%	**24 Jul 2025**	–

Between ① and ③, **49 of 72** days are below 80%: the readings bounce around the floor (daily noise is about ±0.15 points).

Backtest the method: the battery call from four starting points:

Data used	Call made on	Predicted crossing	vs first dip	vs 30-day mean	Warning (to the first dip)
1 year	1 Jan 2022	1 Mar 2025	−74 days	−111 days	1,229 days
2 years	1 Jan 2023	7 Apr 2025	−37 days	−74 days	864 days
3 years	1 Jan 2024	1 May 2025	−13 days	−50 days	499 days
4 years	31 Dec 2024	18 May 2025	+4 days	−33 days	134 days

The original code book says "only the four-year call goes a few days late". True only against the single noisy first dip; against the 30-day mean, **every** call is early.

Why early: a line through each calendar year gives −4.81, −4.69, −4.55, −4.45 and −4.34 % a year. The fade slows, so lines from early data land early. (The pepper task fits a curve, degree 2, on the same three years: **14 Jun 2025**, inside the real window.)

Eclipse ageing: eclipse seasons (27 Feb – 13 Apr and 1 Sep – 16 Oct) are **24.7%** of the days but carry **43.4%** of the battery's fade: **2.34×** faster per day than in sunlight.

The long reach (the line solved directly, fitted on 1–5 years of data):

Data used	Amplifier → −2 dB	Propellant → 5 kg
1 year	27 Nov 2033	23 Oct 2035
2 years	26 Jul 2033	13 Dec 2035
3 years	4 May 2033	6 Dec 2035
4 years	17 Mar 2033	14 Dec 2035
5 years	24 Mar 2033	15 Dec 2035

The amplifier calls span about **8 months**; even the 5-year call reaches **7.2 years** past the last day of data. The propellant calls all sit within **53 days**, because the burns are so regular, but that holds only if the burn plan doesn't change. (The 3-year amplifier call shows as 4 May here and 5 May in A2: the formula rounds the fractional day down, `day_of_threshold` takes the first whole day below. Same line.)

Slide talking points

#	Say this	Ask this
3	"The danfo driver does remaining-useful-life sums every day without the name." The numbers are a story.	"When does the smart driver buy fuel?" (before the light comes on: Quiz Q24)
4	Facts from the toolkit fact sheet only. Our data is a **synthetic** platform.	"Would you bet the 2A/2B dates won't slip?"
6	Three shapes. The limits are known up front.	"Which part would you worry about first?" (the battery)
7	A burn every ~14 days. A straight line through a staircase is fine over years, wrong between two burns.	Quiz Q23
9	Stand on 1 Jan 2024 with three years. Click to reveal what really happened.	"Is the gold line a good forecast?"
10	Six lines of code. `np.polyfit(x, y, 1)` → slope and intercept. Point at `np.argmax`.	Quiz Q20
11	Battery and amplifier look fine… propellant says 1 Jan 2021.	"What went wrong?" (let them guess before slide 12)

#	Say this	Ask this
12	`argmax` of all-False is 0. A confident wrong date with no error.	"Where else could a 'nothing found' turn into a fake answer?"
13	The line does reach 5 kg, just after the search stopped. Solve it directly.	–
14	The fuel-gauge sum agrees. Simple sums catch bugs.	–
16	Daily dots bounce around 80% for 72 days. Three honest definitions.	"Which one would operations care about?"
17	13 days early or 50 days early: same forecast, two scores.	Quiz Q29
18	Four starting points. More data, closer call, less warning.	Quiz Q30
19	The fade slows, so a straight line from early data calls early. Early is the safe side for a battery.	"Would 'late' be safe for a battery?" (no)
20	The battery ages when it works: inverter in a NEPA-heavy month.	–
21	Amplifier: a planning figure with a range, re-estimated every year.	Quiz Q22
22	Five rules, and the model sentences on the right.	Quiz Q24

Answer key

Task	Answer	Common mistake → what to say
A1	`DAYS_PER_YEAR = 365` → −4.59 % a year	Typing 12 (months): the slope is per **day**
A2	`COLUMN = "threshold"` → battery 1 May 2025, amplifier 5 May 2033, propellant `None`	Typing `80` or `threshold` without quotes: the cell asks for a column **name**, as text
A3	`crossed_in_window = False` (and `np.argmax` said day 0)	`True` because "it printed a date": that's exactly the trap
A4	`crossing_day = (5.0 - intercept) / slope` → 6 Dec 2035	Swapping the order `(intercept - 5.0)` : gives a negative day; slope is negative, so keep `(5.0 - intercept)`
A5	`RESERVE_KG = 5.0` → 10.1 years, about January 2036	Typing 0: the reserve is not spare fuel
B1	`FLOOR = 80.0` → 14 May, 20 Jun, 24 Jul 2025	–
B2	`KEY = "30_day_mean"` → (13, 50)	Spaces instead of underscores (KeyError): copy the key from the print

Task	Answer	Common mistake → what to say
B3	`YEARS = [1, 2, 3, 4]` → −74, −37, −13, +4 days; warning 1,229, 864, 499, 134 days	`YEARS = 4` : it must be a list
B4	`"slows"`	"speeds up" because the line goes down faster… read the numbers: the yearly loss gets smaller
C1	`ECLIPSE = 1` → 43.4% of the fade (on 24.7% of the days)	`True` also works (1 == True in Python)
C2	`LAST = -1` → 7.2 years	`0` (that's the first day: gives about 12 years)
C3	`"B"`	`"A"` because it's precise: a precise date 7 years out is false confidence
🌶️	`DEGREE = 2` → 14 Jun 2025, inside the real window	Warn: curves can go wild over long reaches; don't use degree 2 for the amplifier's 9 years

Quiz 4

- **Files:** `2-exercise/quiz/W6D2-quiz-4.pdf` (printable, no answers), `W6D2-quiz-4-google-forms.csv` (import), `4-instructor-only/W6D2-quiz-4-answer-key.pdf`.
- **Source:** the original `Time-series/MCQ/week-06-mcq.csv`, unchanged wording. Options reshuffled with a fixed seed (6203) so the correct letters are spread: multiple choice A ×6, B ×7, C ×6, D ×7; True/False A ×2, B ×2. (In the original file, most answers were option B.)
- **Topic → day:** forecasting baselines → W5D2 · regression-based forecasting → W5D3 · uncertainty and prediction intervals → W5D3 · backtesting → W6D1 (Q29–30 → today) · trend-to-threshold and remaining useful life → today.
- **All 30 answers match the data** (we re-ran every number: MAE 11.25 and 15.25, one-step naive 6.75, residual sd 2.84, held-out error sd about 7, coverage 5 of 8, battery call 1 May 2025, 13 days early, fold MAEs 1.27–7.67 with mean 4.01, naive wins 5 of 6 folds). No contested questions this time. Two notes for marking:
- **Q29** ("the real crossing was 14 May 2025… how far off?", answer **13 days early**) uses the **first-dip** definition of "crossed". The answer stands. Use it to repeat today's point: against the 30-day mean the same call is 50 days early, so always say which definition you used.
- **Q22** ("the amplifier call reaches about 9 years past the data") means past the **three years it was fitted on**: 1 Jan 2024 to 5 May 2033 is 9.3 years. From the end of the whole file (30 Dec 2025) it's about 7 years. Either way: a planning figure with a range.

Troubleshooting

Symptom	Cause	Fix
✅ `Ready!` never appears	No internet in Colab, or the GitHub branch isn't public yet	Runtime → Restart session; check the link opens in a browser

Symptom	Cause	Fix
A participant's dates are one day off a neighbour's	Formula (rounds down) vs `day_of_threshold` (first whole day below)	Same line. Say so and move on
`TypeError` with `NoneType`	They used `day_of_threshold`'s `None` as a number	That's the honest "no crossing": solve the line instead (A4)
The plot cell is slow	Three panels of 1,825 days	Wait a few seconds; it's fine
Someone asks "is this real NigComSat-1R data?"	–	**No.** A synthetic platform from the original code book. Real telemetry stays inside NIGCOMSAT (final-project data rules)

Final-project link

Brief **B (power and battery degradation)** and brief **D (capacity)** use exactly today's method. Their minimum method in the requirements: *a trend-to-threshold estimate giving a crossing date **and a window** (never a single day), with the assumptions stated*, and in the honest test: *the call repeated from several starting points, showing how the date and the warning time change*. Today's Part B table is the template. Next session is **demo day** 🎤 .